

Liars Are Information:

Receiver-Anchored Channel Estimation for Byzantine-Robust Aggregation in Multi-Agent LLM Systems

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Abstract

When some agents in a multi-agent LLM system lie, the classical defence is to discard them. That defence assumes an honest majority and fails at a Byzantine fraction of one half. Prior work observed that a *coherent* liar can instead be read backwards, and gated this inversion on how coherent a channel is. The same work proved that no coherence-based rule survives an adversary that coordinates just below the gate’s threshold. We argue that the informative quantity is not coherence but *dependence on the truth*, and that it can be estimated without labels if the estimator is anchored on the one fact a receiver knows: it is itself honest. We introduce **Race** (Receiver-Anchored Channel Estimation). Each honest agent fits a one-coin latent-truth model to its own unlabelled history, using anchor initialisation, an anchor prior and an anchor constraint. Replicated channels are tempered by error-conditioned clone detection. Every peer receives a signed log-odds weight, which subsumes the trust/discard/invert gate.

We prove that against a receiver that knows the channels a lie can only add information and the minimax attack is to be uninformative, so there is no breakdown point below $f = 1$. Label-free estimators must fail at $f \geq 1/2$ through label switching, which a receiver anchor provably repairs and a near-chance anchor provably cannot. We evaluate on 2,196 replayed worlds (six benchmarks, nine open-weight models, fourteen symbolic attack settings, and four attacks written by LLMs prompted to deceive) and on a live six-model CPU swarm. RACE has no breakdown point in any tested fraction. It scores 95.0% where the gate-aware adversary drives the prior mechanism to 48.5% ($f = 0.7$). Against real LLM deceivers at $f = 0.7$ it scores 89.8%, against 70.8% for the prior mechanism, 29.4% for majority vote and 85.8% for a receiver that trusts only itself. Across 480 Holm-corrected comparisons on that confirmation set it has 196 significant wins and 0 losses. Its label-free channel estimates correlate at $r = 0.977$ with the truth. We report its failure modes as prominently: camouflage attackers that violate conditional independence, sleepers that violate stationarity, and weak receivers that cannot anchor. We also report an audit of the prior artifact. It found an incomplete post-repair recompute and five code defects, and every number here is re-derived from the repaired data.

1 Introduction

Multi-agent LLM systems increasingly decide by pooling: several agents answer, and each agent combines what it hears. Examples include self-consistency [35], debate [11], judge panels [34] and agent swarms [30]. Pooling fails when some agents are adversarial: compromised by prompt injection [12], fine-tuned to deceive [17], or simply instructed to mislead. The classical answer,

*Author list withheld for review. This work extends the “Liars Are Information” artifact of D. J. Das [8] and the derivative release of [28]; code, data, results, figures and videos: see the Reproducibility section.

inherited from Byzantine fault tolerance [4, 25, 36], is to *discard* outliers. That assumes an honest majority and breaks at $f = 1/2$.

The “Liars Are Information” study [8] made a sharp observation: a liar who reliably says X when the truth is not X is not noise. “This agent says X ” is evidence *against* X . Its mechanism, AIP, inverts a channel when the channel is more *coherent* with other dissenters than honest agents ever are. The same study then defeated AIP. An adversary that knows the published coherence ceiling coordinates just below it, and no rule measurable with respect to per-question coherence can separate it from honest disagreement. Two further failures were documented. On binary questions coherence is degenerate, and AIP scores 0%. And on open-answer mathematics AIP fell below a receiver that simply trusts itself.

This paper. We argue that the premise was right and the statistic was wrong. What makes a liar informative is not coherence but *dependence on the truth*. An agent that never states the truth leaks it by elimination, whether or not it coordinates. An agent whose answers are independent of the truth leaks nothing, however coherent it is. Dependence on the truth is exactly what latent-truth models of annotator reliability estimate [9, 37]. They fail here for a different reason: label-free estimation is identified only up to a relabelling of the truth, and with an adversarial plurality it adopts the liars’ labelling. The fix is the one piece of knowledge a receiver always has: *it is itself honest*.

Contributions.

1. **Race (Receiver-Anchored Channel Estimation).** Each honest receiver fits a one-coin latent-truth model to its own unlabelled history. The fit is anchored on the receiver through initialisation, a prior and a constraint. Replicated channels are tempered by error-conditioned clone detection. The resulting signed, continuous log-odds weight subsumes AIP’s TRUST/DISCARD/INVERT gate (Section 4).
2. **Theory.** (i) Against a receiver that knows the channels, a lie can only add information, and the minimax attack is to be uninformative, so there is no breakdown point below $f = 1$ (Theorem 1). (ii) Label-free estimators must fail at $f \geq 1/2$ by label switching; a receiver anchor restores identifiability, and a near-chance anchor provably cannot (Proposition 3). (iii) AIP’s gate is a quantisation of the Bayes weight, and the coherence impossibility does not bind a truth-dependence estimator (Proposition 4). (iv) We give sample-complexity and clone-detection results that explain the design (Propositions 5 and 6).
3. **Evaluation.** Over 2,196 replayed worlds on six benchmarks with nine models, fourteen symbolic attacks and four prompt-injected LLM attacks, RACE holds at every Byzantine fraction up to 0.9. With real LLM deceivers at $f = 0.7$, RACE scores 89.8% against 70.8% for AIP, 29.4% for majority vote and 85.8% for the receiver alone. Against the gate-aware attacker that defeats AIP, RACE scores 95.0% against 48.5%. We report where RACE loses as prominently: a camouflage attacker built against it, sleepers that violate stationarity, and weak receivers that cannot anchor.
4. **A live multi-agent LLM experiment.** Six small open models from six labs answer 240 questions in honest and covert-saboteur roles, and in a debate condition in which honest agents first see a panel that is half liars (??). This is new inference, evaluated after the method was frozen.
5. **An audit and a repaired, re-derived artifact.** We found that the upstream post-repair recompute was incomplete, found four code defects and a replica-cloning defect, and re-derived every number from the repaired cache (Section 9). All code, data, figures, videos and the claims in this paper are regenerated by one script.

2 Related work

Byzantine-robust aggregation. From the Byzantine generals problem [25] to robust distributed learning, the dominant response to adversarial input is to *remove* it. Krum [4], coordinate-wise median and trimmed mean [36], and Bulyan-style composition [27] are examples. These rules assume an honest majority and break at $f = 1/2$. The original “Liars Are Information” study [8] observed that this ceiling belongs to *discarding*, not to the problem. It proposed Adversarial Informativeness Pooling (AIP), which inverts channels whose coherence exceeds a calibrated honest ceiling. The same study then defeated AIP with an adversary that coordinates just below the ceiling, and proved that no coherence-measurable rule can separate such an adversary from honest disagreement. A derivative release [28] added clean-task calibration, clone caps and causal online windows, with mixed results. RACE keeps the premise (a liar can be read backwards) and replaces the coherence gate with channel estimation.

Learning from unreliable annotators. Dawid and Skene [9] introduced per-annotator confusion matrices fitted by EM [10]. Later work added provably optimal spectral initialisation [37], budget-optimal inference [20], joint learning of a classifier [33], and unsupervised ensemble ranking [29]. Identifiability of such latent-class models holds only up to label permutation [1, 23]. Crowdsourcing assumes a better-than-chance *majority*, which is exactly what fails at $f \geq 1/2$. Our contribution is to replace that assumption with a *receiver* anchor (Proposition 3) and to handle replicated channels, which crowdsourcing rarely faces. The “surprisingly popular” algorithm [32] also extracts truth from a minority. It needs a second elicitation (predicted votes) that an adversary could forge, whereas RACE uses only the answers themselves.

Correlated LLM errors. LLMs make correlated mistakes [21]. Judge panels are worth far fewer independent votes than their size suggests [22, 34, 38]. Open-ended outputs are homogeneous across models [18]. Classical jury theorems degrade under correlated votes [24]. These findings motivate our error-conditioned clone detection (Proposition 6). We treat correlation as a quantity to estimate from errors, not from raw agreement.

Multi-agent LLM systems under attack. Sampling-and-voting [35] and multi-agent debate [11] improve reasoning when every agent is honest. Adversarial debaters can pull a group toward wrong answers [2]. Deceptive behaviour can persist in LLMs [17, 31], and agents can be subverted by indirect prompt injection [12]. Our live experiment (??) measures the debate side directly: how exposing honest agents to a lying panel before they commit changes the information available to receiver-local aggregation.

3 Setting and threat model

Swarm. N agents answer a stream of questions $t = 1, 2, \dots$. On each question every agent broadcasts one answer. Receiver i hears its own answer and a subset of its peers’ answers (visibility p_{obs}), and must output a final answer using only what it has heard on this and earlier questions. No labels are ever revealed. A fraction f of the agents is Byzantine; the rest are honest. Every honest agent is a receiver, and we score the mean accuracy over honest receivers. Byzantine receivers are never scored.

Table 1: Attack zoo. “Rushing” attacks see the honest answers before speaking. The last column names the assumption of receiver-anchored estimation that the attack targets. Theorem 1 shows the minimax stationary attack is *uninformative/echo*.

Attack	Behaviour
coherent	one shared wrong answer per question (the original threat)
independent	each attacker lies on its own; never coordinates
gate-aware(p)	shares the lie w.p. p , else lies alone; p tuned below a published ceiling
partial(e)	coherent lie w.p. e , else the truth
uninformative	uniform random answer, independent of the truth
echo	copies a random honest answer ($B \perp Y \mid H$)
attractor	always backs the honest agents’ most common wrong answer (rushing)
camouflage(θ)	truthful unless the honest vote is contested (share $< \theta$), then backs the strongest wrong honest answer (rushing)
sleepers	replays an honest model on history, lies coherently afterwards
LLM: always-wrong, rushing, semantic hallucination, semantic negation	answers written by Llama-3.2-3B / Ministral-3-14B under four deception prompts (replayed)
LLM (live): solo, rushing, collude	six small open models prompted as covert saboteurs (??)

What the receiver knows. One thing: it is itself honest, i.e. it reports its own best answer. It does not know f , who the attackers are, which model produced a peer’s answer, or any label. Self-reported confidence is ignored throughout. It can be forged, and the original study showed a falsified self-report falls below chance [8].

Adversary. Byzantine agents may (i) know the gold answer, (ii) be *rushing*, seeing the honest broadcasts on the current question before speaking, (iii) coordinate arbitrarily with each other, (iv) know the defence and its published constants, and (v) change behaviour over time. Table 1 lists the strategies we evaluate. They are grouped by the assumption of our method each one attacks. Nine are symbolic. Four replay answers that real LLMs wrote when prompted to deceive [8]. Three more (??) come from live inference in this work.

4 Receiver-anchored channel estimation

4.1 Model

For receiver i and question t , let $\mathcal{S}_t$ be the candidate set: the label space for closed questions (C options), or the distinct answers heard on t for open questions. Two open answers are merged when an equivalence checker says they denote the same value. This needs no gold (Section 6). Let $K_t = |\mathcal{S}_t|$. RACE fits the symmetric one-coin latent-truth model

$$Y_t \sim \text{Unif}(\mathcal{S}_t), \quad P(R_{tj} = Y_t) = a_j, \quad P(R_{tj} = c \mid Y_t \neq c) = \frac{1 - a_j}{K_t - 1}, \quad (1)$$

with one accuracy a_j per channel, including the receiver’s own. Channels are conditionally independent given Y_t , up to the clone tempering described below. The posterior score of candidate c is a *signed* weighted vote:

$$s_t(c) = \sum_j w_{tj} \mathbf{1}[R_{tj} = c] \lambda_{tj}, \quad \lambda_{tj} = \log \frac{(K_t - 1) a_j}{1 - a_j}. \quad (2)$$

A report with $a_j > 1/K_t$ is evidence *for* the reported answer (TRUST). A report with $a_j < 1/K_t$ is evidence *against* it, spread over the alternatives (INVERT). $a_j = 1/K_t$ contributes nothing (DISCARD). AIP’s gate is a three-level quantisation of Equation (2). Unlike AIP, RACE decides the sign from a_j , not from how coherent a channel is (Proposition 4).

4.2 Anchored EM

The parameters are fitted by EM [10] on the receiver’s own unlabelled history. Label-free latent-class models are identified only up to a relabelling of the latent truth (Proposition 3). When a coherent bloc holds a plurality, the likelihood prefers the relabelling in which the liars are the experts. RACE removes the ambiguity with the receiver’s self-knowledge, in three ways:

1. **Anchor initialisation.** The first E-step puts posterior mass 0.75 on the receiver’s own answer, not on the plurality.
2. **Anchor prior.** $a_i \sim \text{Beta}$ with mean 0.75 and strength 8. Peers get a weak prior centred at chance (strength 2), so an unfamiliar peer starts at DISCARD, not TRUST.
3. **Anchor constraint.** After every M-step a_i is projected to $\geq 1/\bar{K} + 0.05$, where $1/\bar{K}$ is the chance rate on the tasks the receiver answered.

The M-step is the MAP update $a_j = (\sum_t \omega_t q_t(R_{tj}) + \kappa_j m_j) / (\sum_t \omega_t + \kappa_j)$, where q_t is the current posterior and ω_t an optional task weight (exponential forgetting in the online variant). The E-step is Equation (2) followed by a softmax. Accuracies are clipped to $[10^{-3}, 1 - 10^{-3}]$. At prediction time unseen agents carry zero weight, and ties go to the receiver’s own answer.

4.3 Clone tempering

A swarm usually replicates models. A coordinated bloc is, by construction, many copies of one lie. Under conditional independence both are over-counted. RACE groups channels and tempers each report by $w_{tj} = 1/(\text{number of heard members of } j\text{'s group on } t)$, so a group contributes one unit of evidence per question. Groups are complete-link clusters over two kinds of link:

- **Same-source links:** raw agreement ≥ 0.99 over co-observed history, i.e. essentially identical answer streams. These are applied *before* any fit, so a receiver’s own replica can never be counted as independent confirmation of the receiver.
- **Dependent-error links:** agreement ≥ 0.95 restricted to history tasks where at least one member is wrong relative to a first-stage truth estimate, with at least 8 such tasks.

Raw agreement is the wrong statistic for the second kind: two distinct but accurate models agree on almost every task simply because both are usually right (Proposition 6).

Cost. One EM iteration is $O(|\mathcal{H}| \cdot N \cdot K)$. With 80 history tasks and 10 agents a fit converges in 10–20 iterations and takes a few milliseconds. The live-swarm run (??) spends about three orders of magnitude more time on generating answers than on aggregating them.

Algorithm 1 RACE fit for one receiver i (history $\mathcal{H}$, no labels)

- 1: Encode reports R_{tj} over candidate sets $\mathcal{S}_t$; $K_t \leftarrow |\mathcal{S}_t|$
 - 2: $G \leftarrow$ complete-link groups of same-source links (raw agreement ≥ 0.99)
 - 3: $(a, q) \leftarrow \text{ANCHOREDEM}(R, G)$ $\triangleright$ init at i 's answers; Beta prior; $a_i \geq 1/\bar{K} + \delta$
 - 4: $\hat{y}_t \leftarrow \arg \max_c q_t(c)$
 - 5: $G \leftarrow$ complete-link groups of same-source $\cup$ dependent-error links w.r.t. $\hat{y}$
 - 6: $(a, q) \leftarrow \text{ANCHOREDEM}(R, G)$
 - 7: **return** a, G $\triangleright$ predict with Equation (2), tempering within G
-

Hyper-parameters. All were fixed before any evaluation data were inspected and were never tuned: anchor mean 0.75, anchor strength 8, anchor margin 0.05, peer strength 2, thresholds 0.99/0.95, minimum support 8, clip 10^{-3} . Two design decisions were revised after inspecting development data. Both revisions are recorded in Appendix A, and the superseded variants appear in every table. The method was then frozen, and two confirmation sets that had not been inspected were evaluated (Section 7.3 and ??).

5 Theory

Let Y be the truth on a question, H the honest reports (including the receiver's), and B the Byzantine reports. An adversary strategy σ is any conditional law $B \sim \sigma(\cdot \mid Y, H)$, so the adversary may be gold-aware and rushing.

Theorem 1 (A lie can only add information; the minimax attack is silence). *Let $\text{acc}^*(\sigma) = \mathbb{E}[\max_y P_\sigma(Y=y \mid H, B)]$ be the accuracy of the Bayes receiver that knows the joint law induced by σ , and $\text{acc}_H^* = \mathbb{E}[\max_y P(Y=y \mid H)]$. Then $\text{acc}^*(\sigma) \geq \text{acc}_H^*$ for every σ , with equality whenever $B \perp Y \mid H$. Hence $\inf_\sigma \text{acc}^*(\sigma) = \text{acc}_H^*$, and the infimum is attained, for example by answering at random or by echoing a random honest report.*

Proof. $P(Y=y \mid H) = \mathbb{E}[P(Y=y \mid H, B) \mid H]$. The maximum is convex, so by Jensen $\max_y \mathbb{E}[\cdot \mid H] \leq \mathbb{E}[\max_y \cdot \mid H]$. Take expectations. If $B \perp Y \mid H$, the two posteriors coincide. $\square$

Corollary 2 (No breakdown point for a known-channel receiver). *Since acc_H^* is at least the receiver's own accuracy, the known-channel receiver never falls below "trust yourself", at any $f < 1$. By contrast, any plurality rule reaches 0 against a coherent bloc once $f > 1/2$.*

Theorem 1 is Blackwell's comparison of experiments [3] read adversarially. Its consequence is that robustness is an *estimation* problem. The adversary has exactly three levers:

1. estimation error;
2. violating the model's assumptions, namely conditional independence given Y and stationarity;
3. withholding its own information.

Section 7 is organised around these levers.

Proposition 3 (Label switching, and what the anchor buys). *(a) Binary one-coin model. For Y uniform on $\{0, 1\}$ and conditionally independent reports with accuracies $\theta = (a_1, \dots, a_n)$, the report law under θ equals the law under $\theta' = (1 - a_1, \dots, 1 - a_n)$. If a coherent always-wrong bloc holds a majority, any rule that selects the labelling in which the plurality is accurate outputs the lie on every question. This covers majority vote, plurality-initialised EM and label-free Dawid–Skene. (b) Dawid–Skene. For any permutation τ of latent classes, relabelling the class prior and the rows of*

every confusion matrix by τ leaves the report law unchanged. Uniqueness results [1, 23] identify the model only up to such τ . (c) Anchor. If the receiver’s confusion matrix is strictly row-diagonally dominant, $\pi_i[k, k] > \pi_i[k, l]$ for all $l \neq k$, then among all relabellings of the true parameters only $\tau = \text{id}$ keeps π_i row-diagonally dominant.

Proof of (c). For $\tau \neq \text{id}$ pick k with $m = \tau^{-1}(k) \neq k$. Row k of the relabelled matrix is row m of π_i . Its largest entry is in column $m \neq k$, so dominance fails in row k . $\square$

The anchor is also *necessary*. In the binary case with $a_i = 1/2$, θ and θ' both satisfy any constraint the truth satisfies. A near-chance receiver therefore cannot identify the truth from its own history, whatever estimator it uses. We observe exactly this weak-anchor capture on open-answer questions (Section 8).

Proposition 4 (AIP’s gate quantises the Bayes weight; Proposition 1 of [8] does not bind RACE). *Under the one-coin model a report $R_j = c$ shifts the log-odds of c against any alternative by $\lambda_j = \log \frac{(K-1)a_j}{1-a_j}$, and $\text{sign } \lambda_j = \text{sign}(a_j - 1/K)$. Consider the gate-aware attacker: it joins a shared lie with probability p , otherwise lies independently, and never states the truth. Its pair coherence is $q(p) = p^2 + (1 - p^2)/(K - 1)$, which sweeps $[1/(K - 1), 1]$. Yet $a_j = 0$ and $\lambda_j = -\infty$ for every p .*

Coherence therefore carries no information about the quantity that determines the Bayes weight (??). The original impossibility result concerns rules measurable with respect to per-question coherence. RACE’s statistic depends on the joint law of (R_j, Y) , estimated through the anchored latent model, so it lies outside that class.

Proposition 5 (History needed to invert correctly). *Fix the groups. Let $\hat{y}$ have error rate δ on the n_j history questions where peer j was heard, and let $\hat{a}_j$ be j ’s agreement with $\hat{y}$. Then*

$$P(\text{sign}(\hat{a}_j - 1/K) \neq \text{sign}(a_j - 1/K)) \leq 2 \exp(-2n_j(|a_j - 1/K| - \delta)^2) \quad \text{whenever } |a_j - 1/K| > \delta.$$

Proof. $|\hat{a}_j - \bar{a}_j| \leq \delta$, where $\bar{a}_j$ is the empirical accuracy. Apply Hoeffding’s inequality [15] to $\bar{a}_j$. $\square$

Take an always-wrong liar on four-way questions ($|a_j - 1/4| = 0.25$) and $\delta = 0.1$. Then $n_j = 80$ gives a bound of 0.055, and $n_j = 160$ gives 0.003. Figure 8 measures the realised curve.

Proposition 6 (Raw agreement confuses competence with dependence). *Take two channels that are conditionally independent given Y , with accuracies a, b and symmetric errors. Then $P(R_1=R_2) = ab + \frac{(1-a)(1-b)}{K-1} \rightarrow 1$ as $a, b \rightarrow 1$, while*

$$P(R_1=R_2 \mid R_1 \neq Y \text{ or } R_2 \neq Y) = \frac{(1-a)(1-b)}{(K-1)(1-ab)} \leq \frac{1}{K-1}.$$

For replicas of one source, and for a perfectly coordinated bloc, the second quantity equals 1.

Proof. $P(\text{equal, not both correct}) = (1-a)(1-b)/(K-1)$ and $P(\text{not both correct}) = 1 - ab \geq (1-a)(1-b)$. $\square$

The same confound explains a refuted recipe in the original ledger: “maximise effective swarm size” largely restated roster weakness [8].

6 Experimental protocol

Data. We use the X3-repaired inference cache of the original study [8]: greedy answers of nine open-weight models on six benchmarks. The benchmarks are MMLU [13], MedQA [19] and ARC [6] (four options each), BoolQ [5] (binary), and GSM8K [7] and MATH-500 [14, 26] (open answers). That is 200 questions per benchmark and 500 for GSM8K. The frozen roster has seven models from seven labs, ranging from 44.5% (Llama-3.2-3B) to 94.0% (Gemma-4-31B) on MATH-500. The cache also contains answers that Llama-3.2-3B and Ministral-3-14B wrote under four deception prompts. Before the repair, 944 of 13,500 rows were answers “manufactured” by the extractor from truncated generations (35% of MATH-500 rows). All earlier MATH-500 conclusions are therefore superseded (Section 9).

Worlds. A world has a benchmark, a Byzantine fraction f , an attack, a visibility p_{obs} and a seed. It contains $N = 10$ identities. Each honest identity replays one cached model, round-robin over a shuffled roster, so some models repeat. Byzantine identities are a prefix of one fixed permutation, which nests the corrupted sets across f . Questions are split by a stable hash into disjoint HISTORY / VALIDATION / TEST partitions (40/20/40%). Every receiver fits on HISTORY only and is scored on TEST. VALIDATION is used only to select attacks for the best-response analysis. The defence input is stripped of gold, Byzantine flags, model names and attack labels.

Open answers. Two answers on the same question are merged when the repository’s equivalence checker (symbolic `math_match` for MATH-500, numeric equality for GSM8K) says they denote the same value. This compares answers with *each other*, never with the key, so a deployed receiver can do the same. It is applied identically to every method.

Methods. *Deployable*: receiver alone; majority vote; logprob confidence weighting; SAC filter-refine; AIP gated, trust-only, naive and soft-gate (M1); one-coin label-free Dawid–Skene (plurality-initialised, the unanchored ablation of RACE); full Dawid–Skene [9]; and RACE with its variants (full confusion, self-cap, and three superseded variants kept as ablations). *Oracles* (upper references, not deployable): majority over the honest agents a receiver heard, and a known-channel decoder that uses RACE’s scoring rule with accuracies measured against gold on HISTORY. All AIP variants use the published calibrated ceilings and the fixed code of [28].

Studies.

- **E1 (main):** $f \in \{0, .1, \dots, .9\}$, coherent liars, $p_{\text{obs}} \in \{1, .5\}$, 5 seeds; 600 worlds.
- **E2 (attack zoo):** 14 symbolic attack settings at $f \in \{.3, .5, .7\}$, 3 seeds; 756 worlds.
- **E3 (replayed LLM deceivers, confirmation set):** 4 prompts, $f \in \{.1, .3, .5, .7\}$, 5 seeds; 480 worlds.
- **E4 (composition, replication, swarm size).**
- **E5 (history length).**
- **E6 (online, predict-then-commit, under changing attackers).**
- **E7 (channel-estimation diagnostics).**
- **E8 (live six-model swarm, confirmation set).**

Statistics. Receivers, seeds and peers within a world reuse the same cached answers, so they are not independent samples. We average accuracy over honest receivers and seeds within each question. We then compare methods on the same questions, using a paired percentile bootstrap

Coherent liars: plurality and label-free rules break at $f = 1/2$, AIP at 0.9 (BoolQ: 0.5); RACE holds

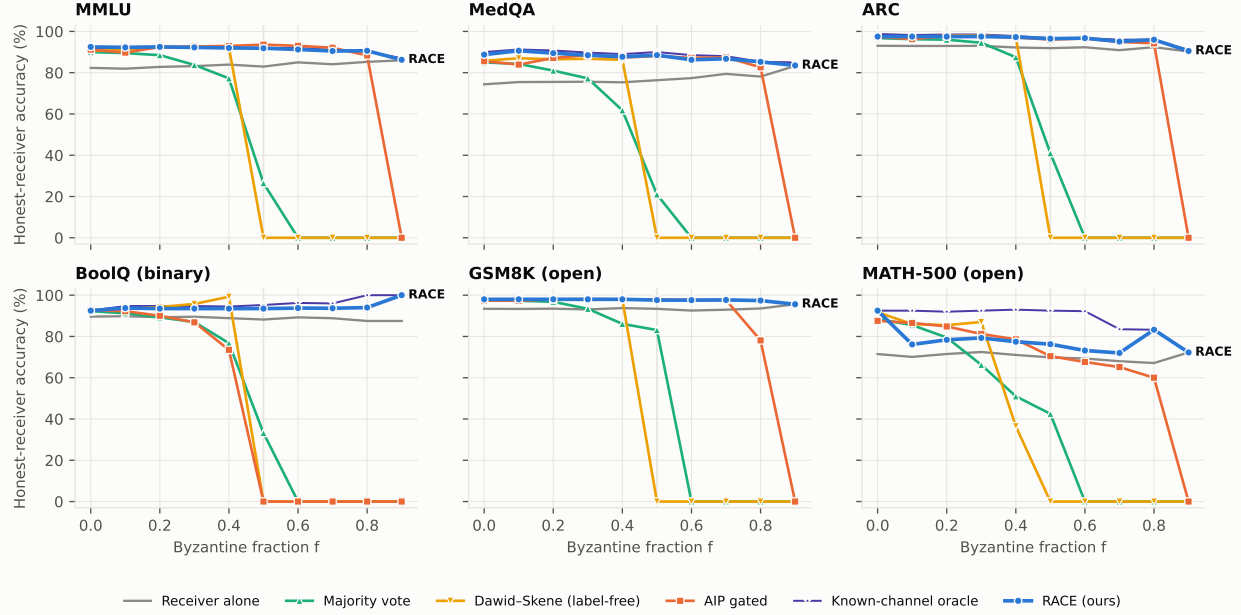


Figure 1: E1. Honest-receiver accuracy against a coherent lying bloc, as a function of the Byzantine fraction f (TEST split, 5 seeds, full visibility). Majority and label-free Dawid–Skene fall to 0 at $f \geq 1/2$ (label switching, Proposition 3). AIP collapses on binary BoolQ at $f = 0.5$ and everywhere at $f = 0.9$. RACE never falls below the receiver alone at any f , except on MATH-500 at small f (weak-anchor capture, Section 8).

(2,000 resamples) for 95% intervals and a Wilcoxon signed-rank test for p -values. Within each study, p -values are Holm-corrected [16] over all cells and baselines. A verdict is “win”/“loss” only if the corrected $p < 0.05$.

Compute. Everything except E8 replays cached answers on a 4-core CPU without a GPU. E8 runs fresh inference with six 1–2B-parameter models on the same CPU. Wall time and peak memory are recorded in each study’s manifest.

7 Results

7.1 E1: no breakdown point against a coherent bloc

Figure 1 is the headline. Averaged over the six benchmarks at $f = 0.5$, RACE reaches 90.7% against 74.3% for AIP, 41.2% for majority vote, 0.0% for label-free Dawid–Skene and 83.8% for the receiver alone. At $f = 0.9$ the figures are 88.0%, 0.0%, 0.0%, 0.0% and 85.8%. The mean accuracy of majority vote and label-free Dawid–Skene first falls below half the receiver’s at $f = 0.5$ and $f = 0.5$. For AIP this happens at $f = 0.9$, and for RACE at none of the tested fractions. The known-channel oracle bounds the remaining headroom at 94.0% ($f = 0.5$).

The binary case, which AIP cannot handle, is where inversion is most valuable. On BoolQ at $f = 0.7$, RACE scores 93.7% (AIP: 0.0%, receiver: 88.8%). A bloc that always says the opposite is a perfect oracle once it is read backwards. With all agents honest ($f = 0$), RACE costs nothing: 93.6% against 92.0% for AIP and 91.7% for majority. Across all 6×10 cells (Holm-corrected Wilcoxon,

Attack zoo (6 benchmarks): blue = pooling helps, red = pooling hurts

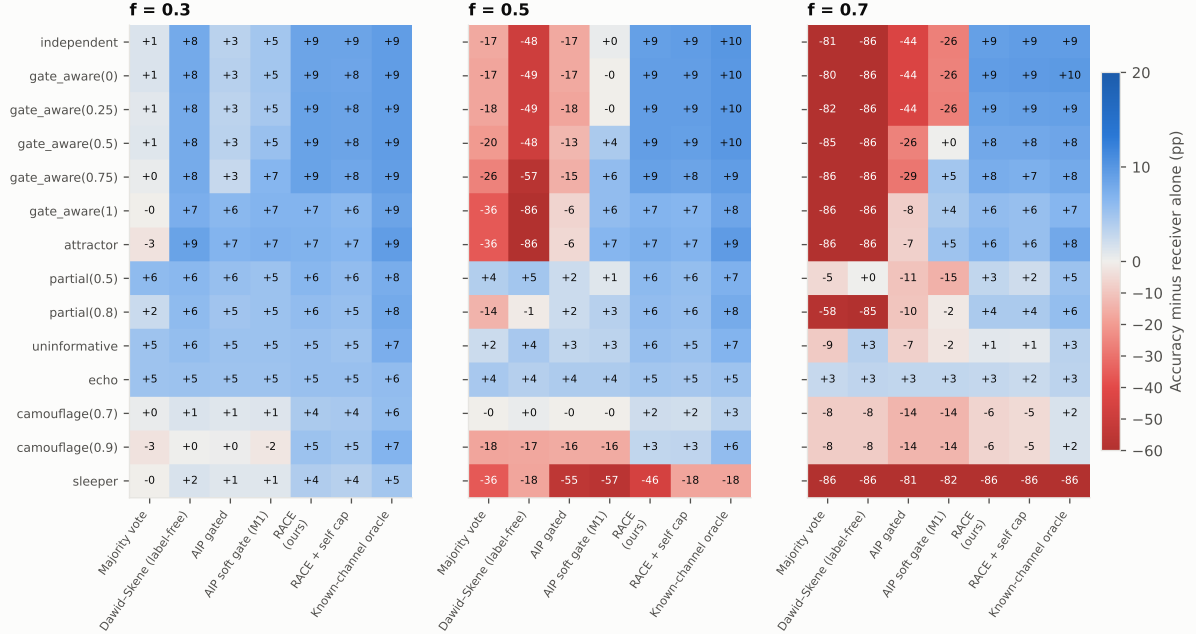


Figure 2: E2. Accuracy minus the receiver alone (percentage points), averaged over six benchmarks and three seeds, for 14 attack settings at three Byzantine fractions. Blue: pooling helps. Red: pooling hurts.

Table 2), RACE is significantly better than AIP in 20 cells and worse in 2. Against majority it wins 41 cells and loses 1. Against the receiver alone it wins 38 and never loses. Every significant loss is on MATH-500 at $f \leq 0.3$, the weak-anchor regime analysed in Section 8. Clone tempering has a price against a *perfectly* coherent bloc. Counting the bloc’s identical vetoes separately happens to help there (93.8% without tempering at $f = 0.5$), but the tempered rule is the one that wins on the confirmation sets (Section 7.3).

7.2 E2: the attack zoo and the gate-aware adversary

Figure 2 organises the attacks by the assumption they target. Three findings stand out.

Coherence gates are blind to incoherent liars. Against the gate-aware attacker at $f = 0.7$ (coordination $p \leq 0.5$, i.e. below AIP’s published ceiling), AIP scores 48.5% and its soft gate 68.9%, while RACE scores 95.0% (Figure 3). Independent liars, who never coordinate at all, give 42.8% versus 95.4%. This is Proposition 4 in data. The attacker’s coherence is tunable, but its dependence on the truth is not.

The minimax attack costs little. Uninformative liars are the minimax strategy of Theorem 1. Against them RACE stays at 87.2% at $f = 0.7$, close to the honest-only oracle (89.5%) and above the receiver alone (86.4%).

RACE’s own failure modes. Camouflage (Section 8) is the one stationary attack that pushes RACE below the receiver at $f = 0.7$ (80.2% vs 86.4%). The sleeper defeats every static fit.

Adaptive attackers. For every defender, benchmark, f and seed we let the attacker pick, on VALIDATION, whichever stationary zoo attack hurts that defender most, and score TEST under that choice (Figure 4). Under its own best response, RACE holds 85.8% at $f = 0.5$ and 75.6% at $f = 0.7$. The corresponding figures are 51.8% and 23.5% for AIP, 42.0% and 0.0% for majority, and

The gate-aware adversary: AIP is defeated below its ceiling; RACE's weight never depended on coherence

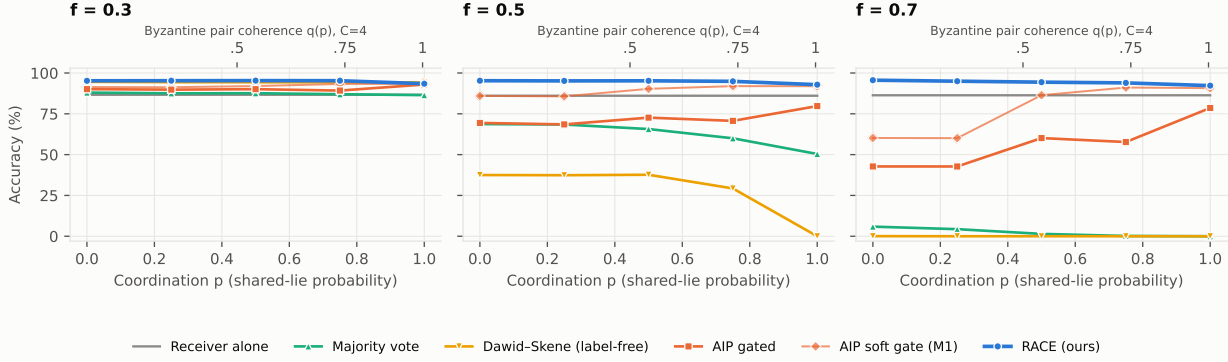


Figure 3: The gate-aware adversary. The upper axis shows the attacker’s pair coherence $q(p)$. AIP is defeated exactly in the sub-ceiling band. RACE is flat in p , because the attacker’s accuracy, not its coherence, sets the weight.

Adaptive attacker: best response per defender, benchmark, f and seed

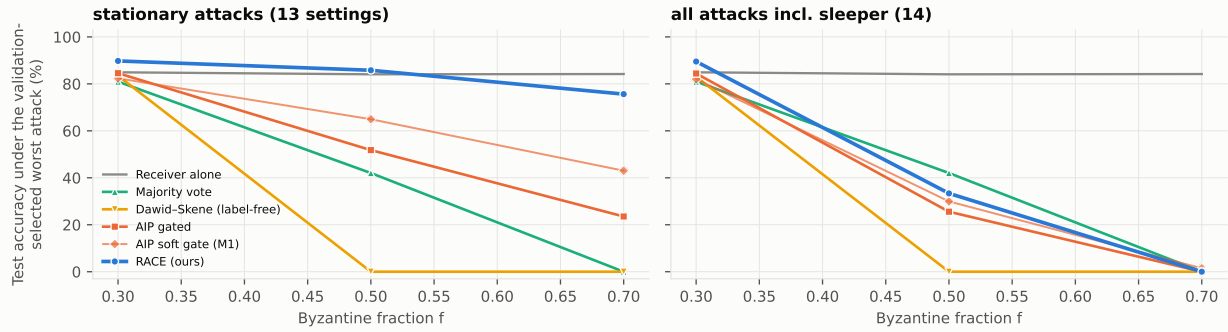


Figure 4: Best-response attacker. The attack is selected on VALIDATION for each defender separately and scored on TEST.

84.1% and 84.2% for the receiver alone. Including the sleeper, every static method fails at $f \geq 0.5$. Section 7.7 addresses this with online refitting.

7.3 E3: replayed LLM deceivers (confirmation set)

The four prompt-injected attacks of the original study were written by real LLMs. Their lies are neither perfectly wrong nor perfectly coordinated. This set was not inspected before the method was frozen. RACE is the best deployable method at every f (Figure 5). At $f = 0.7$ it scores 89.8%, against 70.8% for AIP, 29.4% for majority, 22.7% for Dawid-Skene and 85.8% for the receiver alone. That is within 91.0% of the known-channel oracle and above the honest-only oracle (88.7%), because inverting the liars adds information that honest agents alone do not have. Across 480 Holm-corrected comparisons ($5 \text{ baselines} \times 6 \text{ benchmarks} \times 4 \text{ prompts} \times 4 \text{ fractions}$), RACE has 196 significant wins and 0 losses. Both superseded variants of the clone rule score lower on this set (89.1% and 83.8% at $f = 0.7$). This is independent support for the one post-hoc design change (Appendix A).

Replayed real-LLM deceivers (4 prompts $\times$ 2 attacker models $\times$ 6 benchmarks)

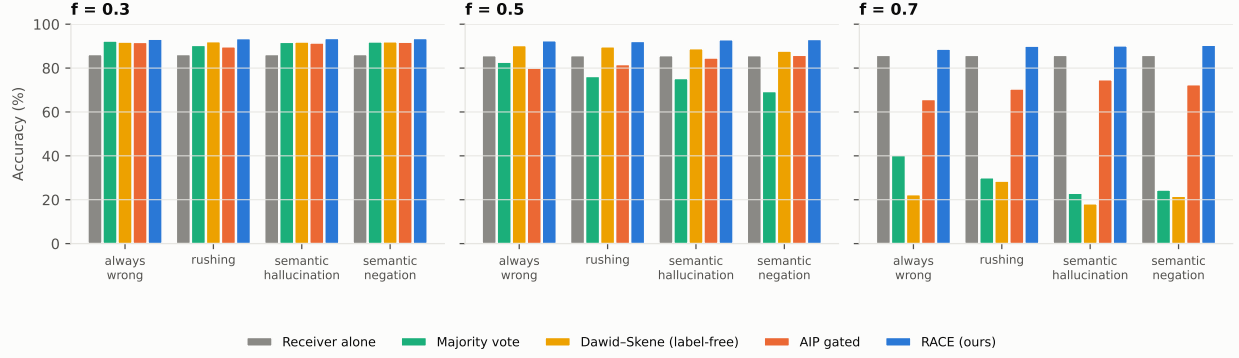


Figure 5: E3. Replayed real-LLM deceivers (Llama-3.2-3B and Ministral-3-14B under four deception prompts). Confirmation set, evaluated after the method was frozen.

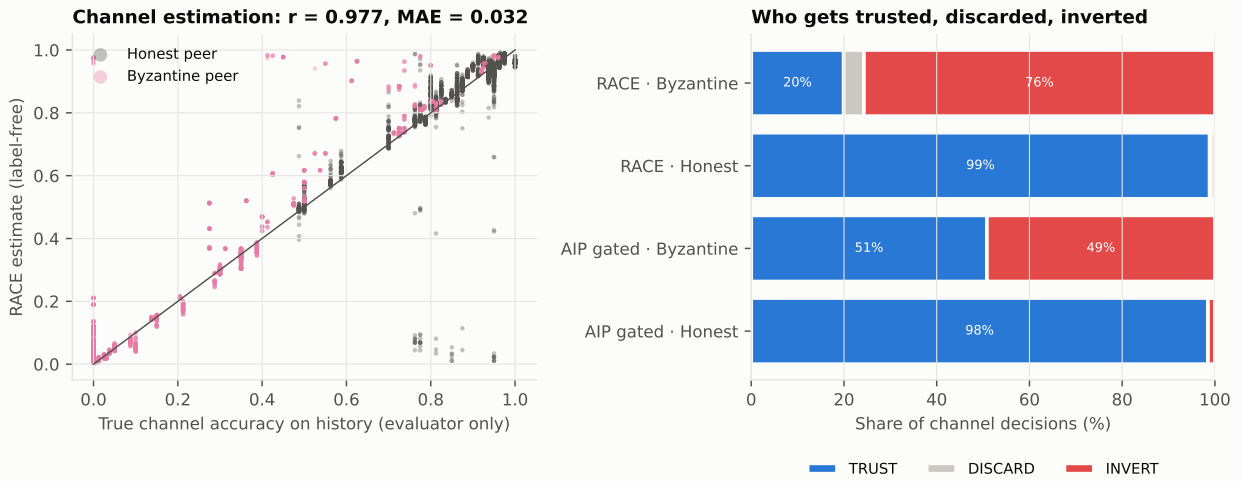


Figure 6: E7. Left: label-free RACE estimates versus true channel accuracy (evaluator-side). Right: share of TRUST/DISCARD/INVERT decisions for honest and Byzantine peers.

7.4 E7: does RACE estimate the channels?

RACE’s estimates are label-free. We compared them with each peer’s true accuracy on HISTORY, which only the evaluator can compute, over 27,216 receiver–peer channels spanning seven attacks, four fractions and six benchmarks (Figure 6). The correlation is $r = 0.977$ and the mean absolute error is 0.032. RACE inverts 75.7% of Byzantine channels, against 49.2% for AIP. It falsely inverts 0.8% of honest channels (AIP: 1.5%). The Byzantine channels RACE trusts (19.9%) are liars that tell the truth often enough to be positively informative, such as partial and camouflage attackers and LLM liars that fail to lie.

Anchor strength. Figure 7 groups receivers by their own accuracy. For receivers above 50%, RACE ends more than 5 points worse than trusting themselves in only 2.1% of cases. The rates are 20.4% for AIP and 38.1% for majority. For receivers at or below 50%, the weak-anchor regime of Proposition 3, the rate rises to 24.1%.

Anchor strength: RACE is almost never worse than self-trust once the receiver is above chance

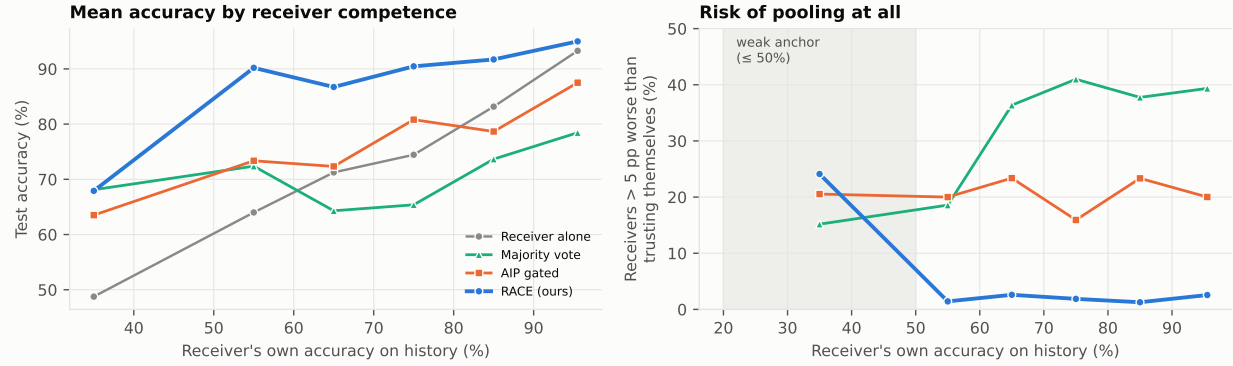


Figure 7: Receiver competence. Left: mean accuracy. Right: share of receivers that end > 5 points worse than trusting themselves.

7.5 E5: how much history?

RACE needs little history (Figure 8). Averaged over attacks and benchmarks at $f = 0.5$, it reaches 85.4% with 5 history questions and 91.0% with 20. AIP reaches 27.9% and 53.9%: its significance test has a minimum-support floor. This matches Proposition 5. An always-wrong liar is far from chance, so few joint observations separate it.

7.6 E4: replication, composition and swarm size

7.7 E6: changing attackers

8 Where RACE fails, and other limitations

Theorem 1 leaves the adversary three levers. We found each one working against RACE.

Violating conditional independence: camouflage. The camouflage attacker tells the truth whenever the honest vote is clear, earning a high estimated accuracy. When the honest vote is contested, it coherently backs the strongest wrong honest answer. Its reports depend on the honest reports, not only on the truth, so the one-coin model misreads it as an expert. At $f = 0.7$ it pushes RACE 6.1 points below the receiver alone. Capping every peer’s accuracy at the receiver’s own (RACE+self-cap) recovers part of the loss. No deployable method we tested beats the receiver alone under this attack at $f = 0.7$. Conditioning the channel model on the honest vote pattern is the natural next step.

Violating stationarity: sleepers. A sleeper that behaves honestly on history and lies afterwards defeats every static fit, including the known-channel oracle, which is fitted on the same history (Figure 2). The causal online study (Section 7.7) shows that refitting with forgetting recovers, with a lag set by the forgetting rate.

Weak anchors. Identification requires the receiver to be better than chance on the *effective* answer space (Proposition 3). On open-answer questions the candidate set is often just {consensus, one alternative}. There a receiver near 50% is effectively at chance. A rushing attacker that lies

How much history does inversion need? (MMLU, MedQA, BoolQ, MATH-500)

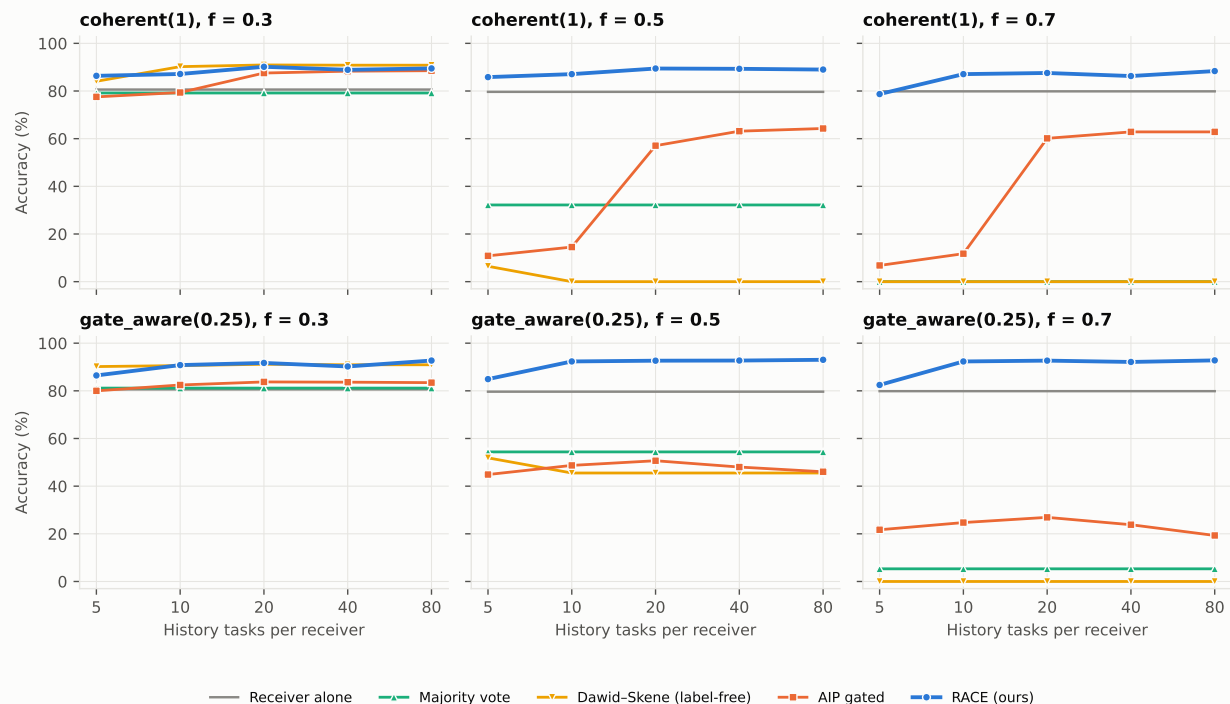


Figure 8: E5. Accuracy versus history length.

with that receiver’s own wrong answers then produces a labelling that is observationally equivalent to the truth from that receiver’s viewpoint. In our MATH-500 runs this affects the weakest roster member (Llama-3.2-3B, 44.5% on MATH-500). Stronger receivers in the same swarm are unaffected. Figure 7 quantifies the gain as a function of receiver competence. It is also why RACE trails AIP and majority on MATH-500 at small f .

Scope. All replay studies use cached single-turn answers. Receivers aggregate final answers, not reasoning, and the symbolic attackers are gold-aware by design. The live study (??) uses small 1–2B models on a CPU and 120 questions per benchmark, so it is a confirmation, not a benchmark. Open-answer scoring relies on an approximate symbolic equivalence checker. Correlated honest errors (shared attractors) remain partly unmodelled: clone tempering only removes near-duplicates. The known-channel oracle shows the remaining headroom.

9 Audit of the prior artifact

A journal-grade result on cached model outputs is only as sound as the cache and the code that reads it. Before building on the prior releases [8, 28] we audited them. Every item below was verified by running code or diffing files.

Truncation (X3). In the original cache, 944 of 13,500 answers were *manufactured*: a 640/2048-token budget truncated completions, and the extractor then read a number out of the unfinished working, at a 100% parse rate. The upstream repair regenerated 2,313 rows. Our loader reproduces the repaired single-model table exactly. All our results use the repaired cache. MATH-500 con-

Table 2: Significant wins/ties/losses of RACE against each baseline (Holm-corrected Wilcoxon signed-rank, $\alpha = 0.05$, task as unit).

Study	verdict Baseline	Wins	Ties	Losses	Cells
E3 LLM deceivers	AIP gated	37	59	0	96
	AIP soft gate (M1)	28	68	0	96
	Dawid-Skene (label-free)	35	61	0	96
	Majority vote	37	59	0	96
	Receiver alone	59	37	0	96
E1 coherent sweep	AIP gated	20	38	2	60
	AIP soft gate (M1)	14	44	2	60
	Dawid-Skene (label-free)	31	26	3	60
	Majority vote	41	18	1	60
	Receiver alone	38	22	0	60
E2 attack zoo	AIP gated	100	147	5	252
	AIP soft gate (M1)	73	175	4	252
	Dawid-Skene (label-free)	87	157	8	252
	Majority vote	122	127	3	252
	Receiver alone	136	96	20	252

clusions in the earlier derivative release, which predates the repair, are superseded. For example, three strong models gained 26–50 points on MATH-500.

Incomplete recompute. After the repair, four result files that the manuscript reads (`n_scaling`, `minimax_regret`, `accuracy_landscape`, `gate_aware_mitigations`) are byte-identical to their pre-repair versions. All 54,432 frozen-roster rows of the aggregation sweep are value-identical, because the sweep resumed and skipped existing cells. The headline macros of the upstream manuscript are therefore pre-repair. A fresh recompute of the gate-aware anchor gives 0.087/0.889 where the paper prints 0.082/0.878. We re-derive every number from scratch.

Code defects. With $p_{\text{obs}} < 1$ the upstream AIP fit paired answers from *different* questions. It seeded the randomised threshold with Python’s salted `hash()`, so runs were not reproducible across processes. It kept stale state across refits, and it counted missing answers as disagreements. The derivative release fixed all four. We build on the fixed code, and all our seeds are SHA-256-derived.

Nondeterminism. Regenerated Llama-3.2-3B rows flip their extracted answer in 52 of 300 re-runs (17.3%) at identical settings. We report this and treat the regenerated rows as fresh samples.

Pipeline extension (MX). Same-model “replicas” were byte-identical because one seed was shared across the batch, so the negative controls passed trivially. The extension stopped before any liar ran. Our live experiment uses distinct models and explicit roles instead.

None of these defects changes the qualitative picture of AIP (inversion helps against a coherent bloc; the gate-aware attacker defeats it). All of them change numbers. The repaired, re-derived numbers are the ones reported here.

10 Conclusion

“Liars are information” is true in a precise sense. A lie is information exactly to the extent that it depends on the truth, and a receiver that knows it is honest can measure that dependence without labels. Replacing a coherence gate with receiver-anchored channel estimation removes the binary collapse and the gate-aware defeat of the prior mechanism. Across all six benchmarks and Byzantine

fractions up to 0.9, it is never significantly worse than a receiver that trusts only itself, with three exceptions: the camouflage and sleeper attacks, and one open-answer cell where the receiver cannot anchor. It also holds up on real LLM deceivers that were not inspected during development. The same analysis says where the idea must stop. An attacker that answers independently of the truth can only silence itself. An attacker whose lies depend on the honest votes, or change over time, attacks the estimator’s assumptions rather than its weights. A receiver at chance cannot anchor anything. Two directions follow. First, channel models conditioned on the honest vote pattern would address camouflage. Second, protocols that collect independent answers before agents see each other would preserve the information that pooling needs (??).

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A Development log: changes made after inspecting data

A change was admissible only if all three of the following held:

1. it fixes a mechanism that can be stated without reference to a benchmark number;
2. it keeps every previously fixed hyper-parameter;
3. the superseded variant stays in every table.
1. **Initial design.** Anchored one-coin EM with clone tempering by raw agreement ≥ 0.95 (kept as `race_rawclone`).
2. **Rejected: multi-start MAP.** We ran EM from both the receiver’s answers and the plurality and kept the higher log-posterior (`race_ms`). Trigger: a near-chance MATH-500 receiver captured by a rushing liar. It did not help, because with a near-chance anchor the two labellings are likelihood-equivalent. It was harmful when replicated channels inflate the likelihood of the adversarial labelling. The anchor therefore acts through initialisation and constraint, not through a likelihood comparison.
3. **Adopted: two-part clone rule.** Same-source links (raw agreement ≥ 0.99 , applied before any fit) and dependent-error links (agreement ≥ 0.95 on tasks where a member errs). Trigger: GSM8K with uninformative/partial liars at $f = 0.7$, where raw-agreement grouping merged distinct accurate models and collapsed honest evidence (Proposition 6).

The method was then frozen, and every study was re-run from scratch. The replayed LLM deceivers (E3) and the live swarm (E8) had not been inspected at freeze time. On E3 the adopted rule outperforms both superseded variants at every $f \geq 0.5$.

B Hyper-parameters and implementation

Anchor initial posterior on own answer	0.75
Anchor prior (receiver accuracy)	Beta, mean 0.75, strength 8
Anchor constraint	$a_i \geq 1/\overline{K} + 0.05$
Peer prior	Beta, mean = chance, strength 2
Same-source / dependent-error thresholds	0.99 / 0.95
Minimum support for a link	8 tasks
Accuracy clip	$[10^{-3}, 1 - 10^{-3}]$
EM	at most 200 iterations, tolerance 10^{-6}
Decision band for reporting TRUST/INVERT	$ \lambda > 0.25$ (diagnostic only)
Online forgetting (E6)	$\gamma = 0.97$, refit every 8 tasks

C Reproducibility

All code, data, results, figures, videos and this manuscript are in the accompanying repository and Hugging Face release. `experiments/run_all.sh` runs E1–E6 on a 4-core CPU without a GPU. `experiments/live_swarm.py` regenerates E8. `experiments/make_figures.py` and `make_numbers.py` regenerate every figure, table and number in the text: the numbers in this paper are macros written by that script. Each study directory holds a `run_manifest.json` with the input and code hashes, wall time and peak memory. The repository’s test suite covers RACE’s identification, invariance, clone-grouping and causality properties.